# EXPERIMENT – 04

**Aim:** Design and train a convolutional neural network for Image Classification for domain of your choice.

**Theory:**

**Introduction to Image Classification**

Image classification is a fundamental task in computer vision that involves assigning a label to an input image from a predefined set of categories. The goal is to train a model that can recognize patterns, textures, and features in images to accurately classify them. Traditional methods relied on handcrafted feature extraction techniques, but deep learning, especially Convolutional Neural Networks (CNNs), has revolutionized image classification by learning features automatically from raw pixel data.

**Convolutional Neural Networks (CNNs)**

CNNs are a class of deep neural networks specifically designed for processing structured grid data such as images. They are highly effective in capturing spatial and hierarchical features by applying learnable filters to image data. CNNs consist of multiple layers, including convolutional layers, pooling layers, fully connected layers, and activation functions.

**a) Layers in a CNN**

- **Convolutional Layer:**
  - The primary feature extraction layer that applies a set of filters (kernels) to the input image.
  - Each filter detects different patterns like edges, textures, or complex features.
  - Convolution operation preserves the spatial structure of the image.
- **Activation Function (ReLU):**
  - The Rectified Linear Unit (ReLU) activation function introduces non-linearity, helping the network learn complex relationships.
- **Pooling Layer:**
  - Downsamples the feature maps to reduce spatial dimensions and computational complexity.

- Commonly used pooling techniques include max pooling and average pooling.
- Max pooling retains the most prominent features, enhancing robustness to small spatial variations.

- **Flatten Layer:**
  - Converts the 2D feature maps into a 1D vector to be processed by fully connected layers.
- **Fully Connected Layer (Dense Layer):**
  - Connects all neurons from the previous layer to extract high-level features and perform classification.
  - The final dense layer has an output equal to the number of classes and uses the softmax activation function for multi-class classification.
- **Dropout Layer (Regularization):**
  - Prevents overfitting by randomly deactivating neurons during training, forcing the model to generalize better.

**Dataset: Fashion MNIST**

The Fashion MNIST dataset, developed by Zalando, is a widely used benchmark dataset for image classification tasks. It consists of:

- 60,000 grayscale images for training and 10,000 for testing.
- Each image is 28×28 pixels, representing a fashion item (e.g., T-shirt, dress, sneaker).
- 10 classes: T-shirt, Trouser, Pullover, Dress, Coat, Sandal, Shirt, Sneaker, Bag, Ankle Boot.

**Model Training and Evaluation**

The CNN model is compiled using the Adam optimizer, which is an adaptive learning rate optimization algorithm. The loss function used is categorical cross-entropy, suitable for multi-class classification problems. The model is trained for multiple epochs with batch processing to improve accuracy.

**Evaluation Metrics:**

- **Accuracy:** Measures the percentage of correctly classified images.
- **Loss:** Represents the error between predicted and actual labels.

- **Validation Accuracy and Loss:** Used to monitor the model's performance on unseen data.

**Results and Predictions**

After training, the model is evaluated on the test set, and accuracy is measured. Predictions are made on test images, and the output is visualized to check the correctness of classifications.

The designed CNN successfully classifies fashion items with high accuracy. By leveraging convolutional layers for feature extraction and dense layers for classification, the model achieves significant performance improvements compared to trad. machine learning techniques.